

Feature engineering

Module 3 - Lesson 3

Overview

Features are the inputs a model learns from. Feature engineering transforms raw data into predictors that make patterns easy to learn.

Key Concepts

- Clean features matter more than clever algorithms; good features beat bigger models.
- Create derived features: ratios, differences, aggregations, and date parts.
- Encode categorical data with one-hot encoding or label encoding.
- Scale numeric features for distance-based models with StandardScaler.
- Missing values become features: a missing flag can itself be predictive.
- Select features by importance and domain knowledge; remove noise.

Example

```
df["revenue_per_hour"] = df["revenue"] / df["hours"]
df["signup_weekday"] = df["signup_date"].dt.dayofweek
df["is_weekend"] = df["signup_weekday"].isin([5, 6]).astype(int)

df = pd.get_dummies(df, columns=["plan"], prefix="plan")
```

Summary

Feature engineering converts raw data into learnable predictors. Derived ratios, proper encoding, scaling, and informative missing-value flags often improve models more than algorithm choice.

Practice Checklist

- Create three derived features from a dataset and justify each.
- One-hot encode a categorical column and inspect the result.
- Compare model performance with and without your new features.